

A Comprehensive Measure of Well-Being

Project Documentation

This project presents an interactive web-based application that predicts a country's Human Development Index (HDI) category by evaluating three core dimensions: health, education, and income. The goal is to demonstrate how development can be measured beyond economic growth alone and how a simplified model can classify countries into Very High, High, Medium, or Low human development categories.

1. Introduction

The Human Development Index (HDI) is a composite measure introduced to emphasize that development should be assessed through people's capabilities and quality of life rather than only economic performance. In this project, the app translates this idea into a practical digital tool that takes key indicators as input and returns a development category.

2. Objectives

- To build a user-friendly web application for HDI prediction.
- To demonstrate how life expectancy, education, and income contribute to human development.
- To provide scenario-based examples for Very High, Medium, and Low development levels.
- To make the concept accessible through a simple and interactive interface.

3. Methodology

The application uses a simplified scoring approach inspired by HDI principles. It calculates separate health, education, and income scores and combines them into a single composite score. The resulting score is then mapped to an HDI category.

Health score: based on life expectancy.

Education score: derived from mean years of schooling and expected years of schooling.

Income score: based on GNI per capita using a logarithmic scale to reflect the diminishing effect of very high incomes.

4. System Architecture

The system consists of a lightweight Python backend and a frontend interface. The backend serves the HTML page and processes prediction requests, while the frontend collects user input and displays the predicted HDI category.

Frontend components: index.html, styles.css, script.js

Backend component: app.py

5. Features of the Application

- Interactive input form for life expectancy, schooling, and income data.
- Three scenario buttons representing common HDI development profiles.
- Instant prediction output showing the category and composite score.
- Readable visual design that makes the model easy to understand.

6. Scenario Examples

Scenario 1: A country with high life expectancy, strong education, and high income produces a Very High HDI prediction.

Scenario 2: A mid-range profile results in a Medium HDI category, indicating room for improvement.

Scenario 3: A low-income, low-education, and low-life-expectancy profile produces a Low HDI prediction.

7. How to Run the Project

1. Open the project folder in the terminal.
2. Run the command: `python3 app.py`
3. Open the local address: `http://127.0.0.1:8000/`

8. Conclusion

This project successfully demonstrates how human development can be modeled in a simple, educational, and accessible way. It combines real-world development concepts with web technology to provide an interactive experience for understanding HDI.